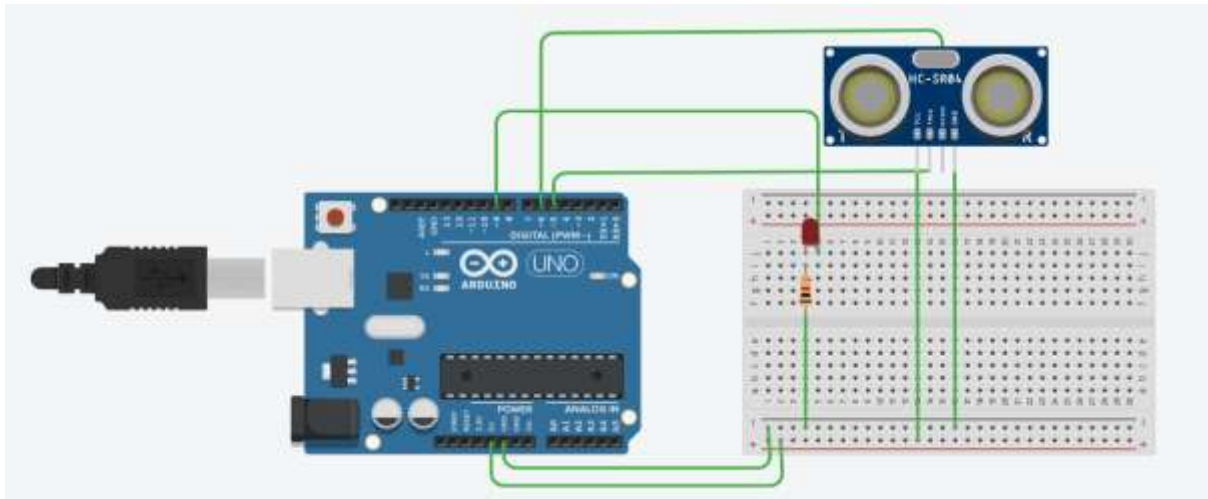




```
const int trigPin = 5;

const int echoPin = 6;

const int ledPin = 9;

const float minDistance = 20.0;

const float maxDistance = 200.0;

float customMap(float input, float inputMin, float inputMax, float outputMin, float
outputMax);

{

    float output;

    output = outputMin + ((input - inputMin) * (outputMax - outputMin)) / (inputMax -
inputMin);

    return output;

}

void setup()

{

    pinMode(trigPin, OUTPUT);

    pinMode(echoPin, INPUT);

    pinMode(ledPin, OUTPUT);

    Serial.begin(9600);

}
```

```
void loop()
{
    long duration;

    float distance;

    float brightness;

    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);
    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin, LOW);
    duration = pulseIn(echoPin, HIGH);
    distance = (duration * 0.0343) / 2.0;
    if (distance <= minDistance)
    {
        brightness = 0;
    }
    else if (distance >= maxDistance)
    {
        brightness = 255;
    }
    else
    {
        brightness = customMap(
            distance, minDistance, maxDistance, 0.0, 255.0);
    }
    analogWrite(ledPin, (int)brightness);
    Serial.print("Distance: ");
```

```

Serial.print(distance);

Serial.print(" cm");

Serial.print("  Brightness: ");

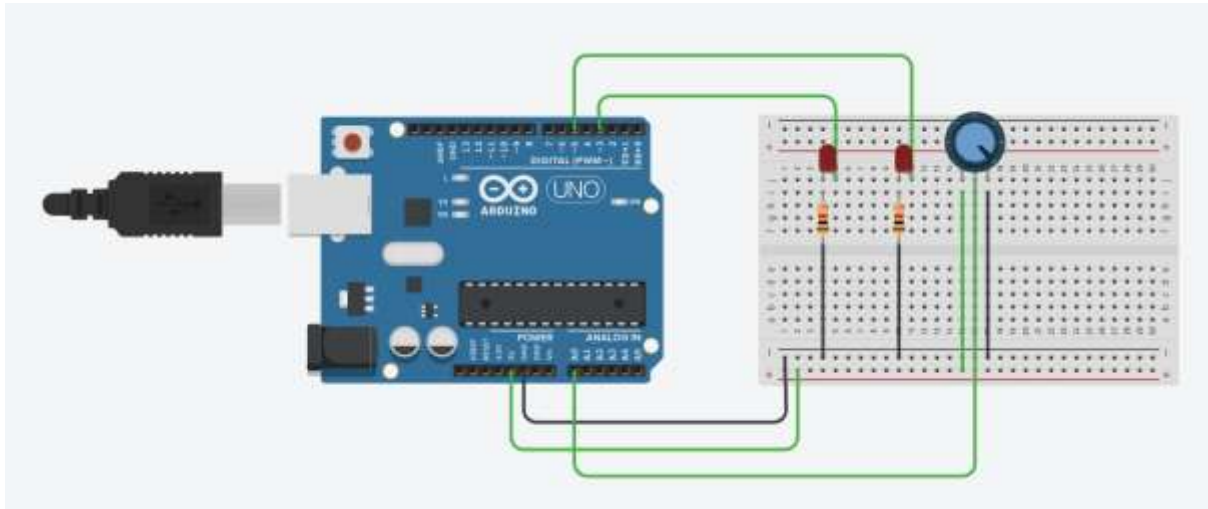
Serial.println(brightness);

delay(50);

}

2)

```



```

int potPin = A0;

int led1 = 3;

int led2 = 5;

void setup() {
  pinMode(led1, OUTPUT);
  pinMode(led2, OUTPUT);
}

void loop() {
  int potValue = analogRead(potPin);
  int brightness1 = map(potValue, 0, 1023, 0, 255);

```

```
int brightness2 = 255 - brightness1;  
analogWrite(led1, brightness1);  
analogWrite(led2, brightness2);  
delay(10);  
}
```